

# **Development Of SmartHire:**

## **AI-Powered Smart Job Search Management System**

Othi Khan

A Practicum in the Partial Fulfillment of the Requirements for the Award of Bachelor of  
Computer Science and Engineering (BCSE)  
The practicum has been examined and approved,

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IUBAT—International University of Business Agriculture and Technology

Summer 2026

## Letter of Transmittal

15<sup>th</sup> September , 2026

The Chair

Practicum Defense Committee

Department of Computer Science and Engineering

IUBAT – International University of Business Agriculture and Technology

4 Embankment Drive Road, Sector- 10, Uttara Model Town

Dhaka-1230, Bangladesh

Subject: Letter of Transmittal,

Dear Sir,

With due respect, I am submitting my practicum report titled “**Development of an AI-Powered Smart Job Search Management System**” as part of the requirements for the Practicum course in the Bachelor of Computer Science and Engineering program. The project was developed during my internship in Web Development at Touch & Solve Technology Ltd.

It was a great opportunity for me to work on this project and apply my theoretical knowledge in a practical environment. During my internship, I gained valuable experience in software development as well as significant exposure to the professional and business culture of an organization.

I sincerely look forward to your valuable comments and suggestions regarding this report. I would be very grateful if you kindly take the time to review this report and evaluate my performance.

Yours sincerely,

---

Othi Khan

Student ID: 22303374

## **Student's Declaration**

I, Othi Khan, a student of the Bachelor of Computer Science and Engineering program at International University of Business Agriculture and Technology, declare that this practicum report titled “Development of an AI-Powered Smart Job Search Management System” has been prepared for the academic requirements of my practicum course.

The report describes the SmartHire system developed during my Web Development internship at Touch & Solve Technologies Ltd. The implementation is based on an ASP.NET Core MVC application using C#, Entity Framework Core, SQL Server, ASP.NET Core Identity, Razor Views, HTML, CSS, and Bootstrap. The report has been prepared from the implemented project, development work, and supporting project documentation.

I further declare that, to the best of my knowledge, the work presented in this report has not been submitted elsewhere for another academic degree, diploma, or certification.

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**Othi Khan**

Student ID: 22303374

## **Supervisor's Certification**

This is to ensure that the Practicum report on the “ **AI-Powered Job Search Management System**” is compiled by Othi Khan, with ID-22303374, of

IUBAT–International University of Business Agriculture and Technology, as part of the fulfillment of the required part of an effective defense course. The report has been prepared under my supervision and is a record of the work accomplished, successfully completed. To the best of my knowledge and as per his declaration, no portions of this report have been posted anywhere by any degree, diploma or certificate.

You are now allowed to submit a report. I wish her every success in her future endeavors. Practicum Supervisor

---

Md. Nazir Ahmed

Lecturer

Department of Computer Science and Engineering

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# **Chapter 1**

## **Introduction**

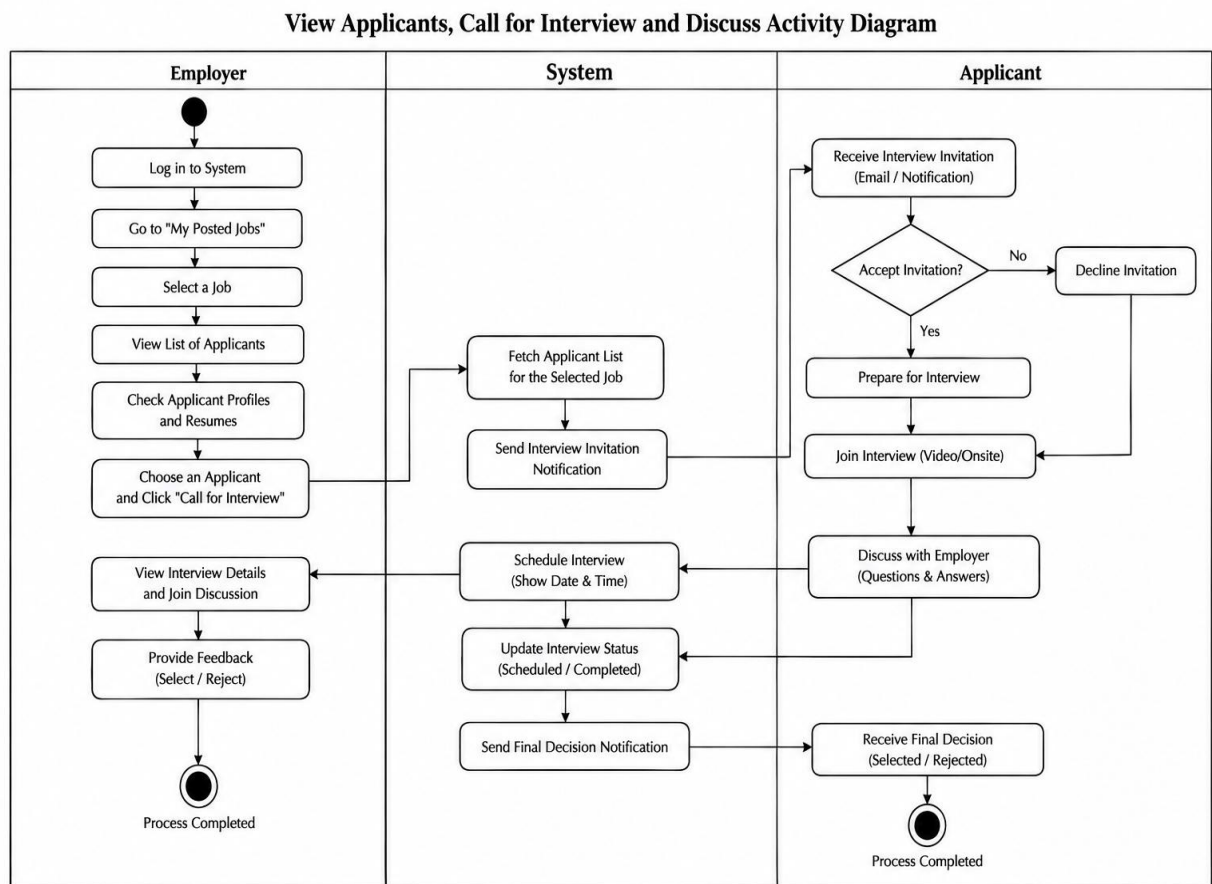
## **Chapter 3**

# **Requirement Engineering**

## 6.4 Swim Lane Diagrams

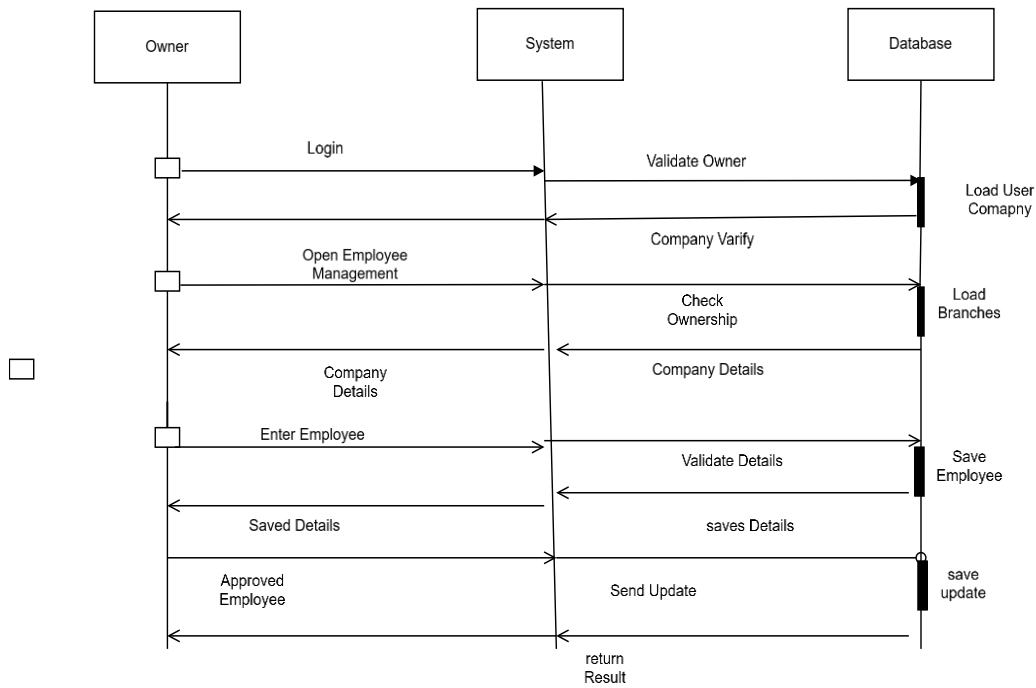
A Swim Lane Diagram is a graphical representation of a process in which activities are divided into separate lanes according to the person, role, department, or system component responsible for performing them. Swim lane diagrams are particularly useful for SmartHire because several operations require cooperation between users and the system. For example, a job-posting workflow involves the employer, application logic, database, and job seeker. Similarly, an application workflow involves the job seeker, system, database, and employer.

The main purpose of a swim lane diagram is to make responsibility clear and show where information is transferred between participants. The reference report also describes swim lanes as a method for identifying responsibility, communication, handoffs, and possible process bottlenecks.

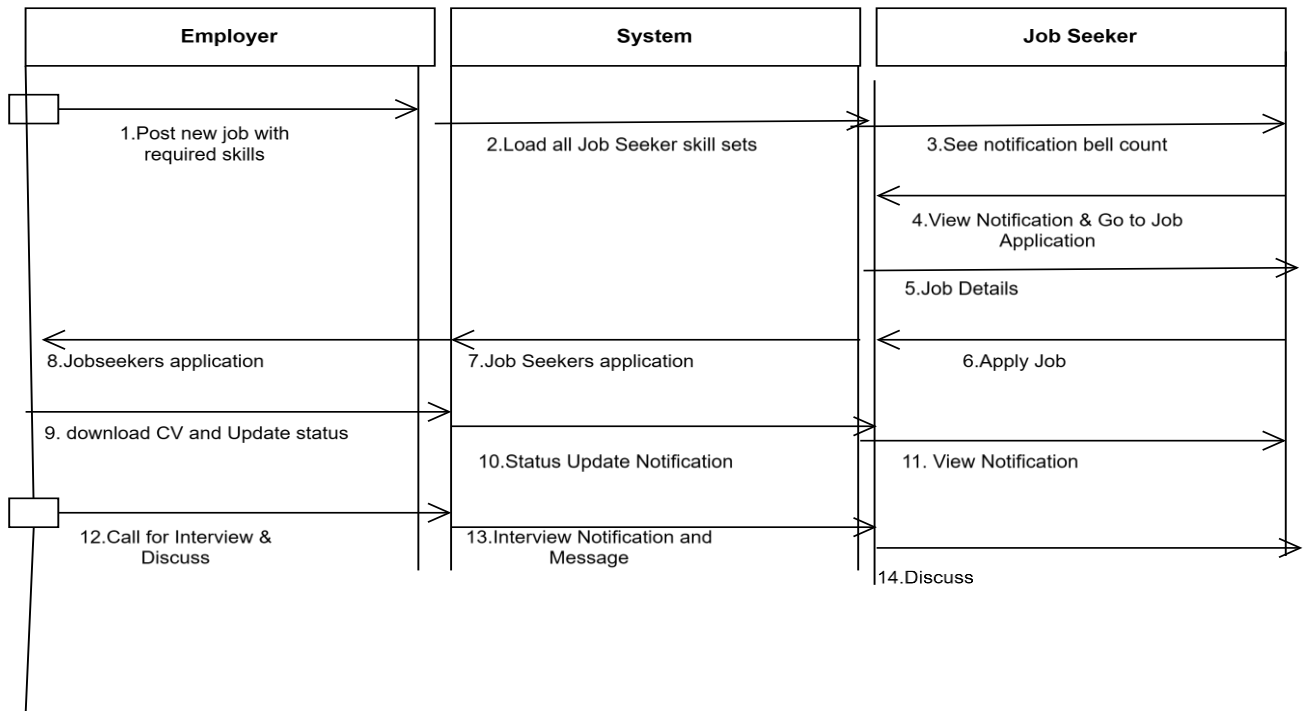


**Figure 6.6: Swim Lane Diagram for View Application and Discussion**

**Figure 6.7 – Swim Lane: Company Owner Adds Employee**

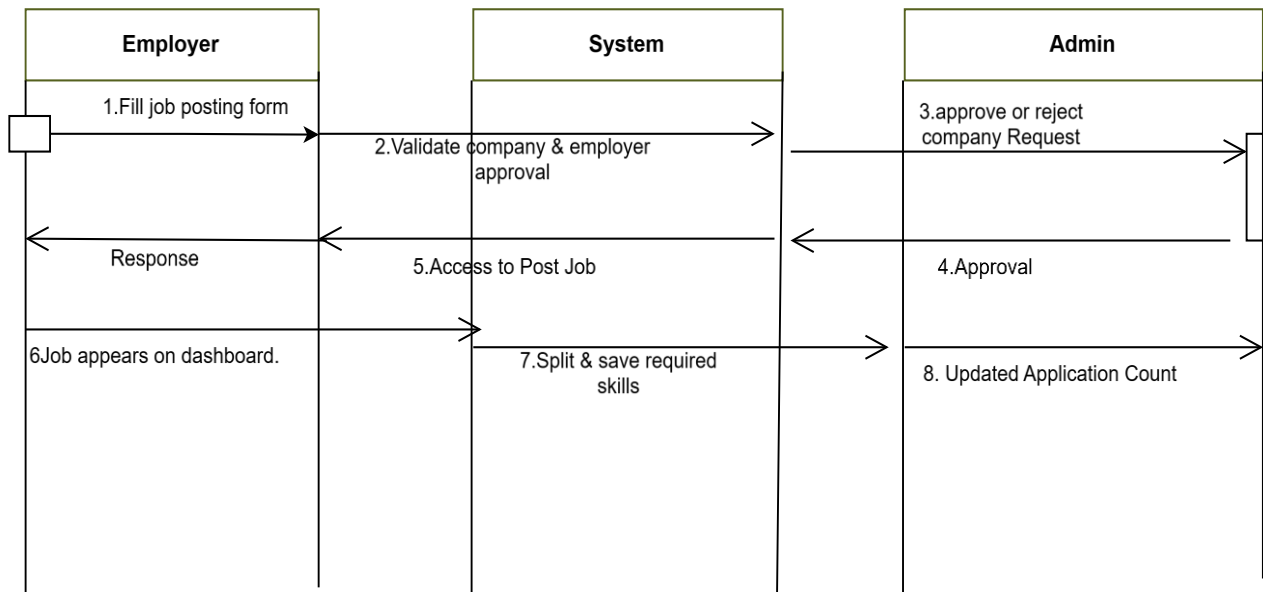


**Figure 6.7: Swim Lane Diagram for Company owner add employee**



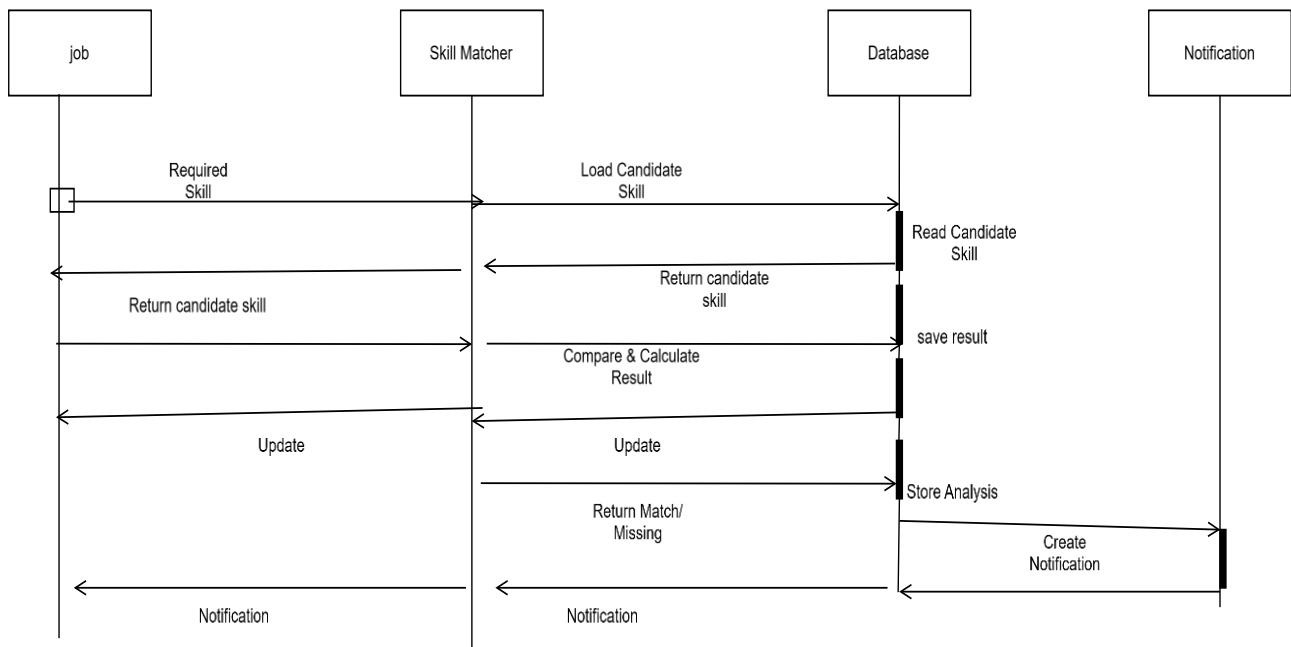
**Figure 6.8: Swim Lane Diagram for Job-Seeker Application**

## Swimlane Diagram for Job Posting



**Figure 6.9: Swim Lane Diagram for Job Posting**

**Figure 6.10 – Swim Lane: Skill Matching and Notification**



### 6.5.2 Level-1 Data Flow Diagram

The Level-1 DFD divides SmartHire into the following major processes:

1. Authentication : registration, login, logout, identity, and role access.
2. Profile Management : profiles, resumes, and skills.
3. Company and Job Management : companies, branches, employers, jobs, and required skills.
4. Application and Matching : applications, resumes, and skill comparison.
5. Notification : notification creation, storage, and display.
6. Administration : company approval, user management, and system control.

#### 6.4.2 Level-1 DFD

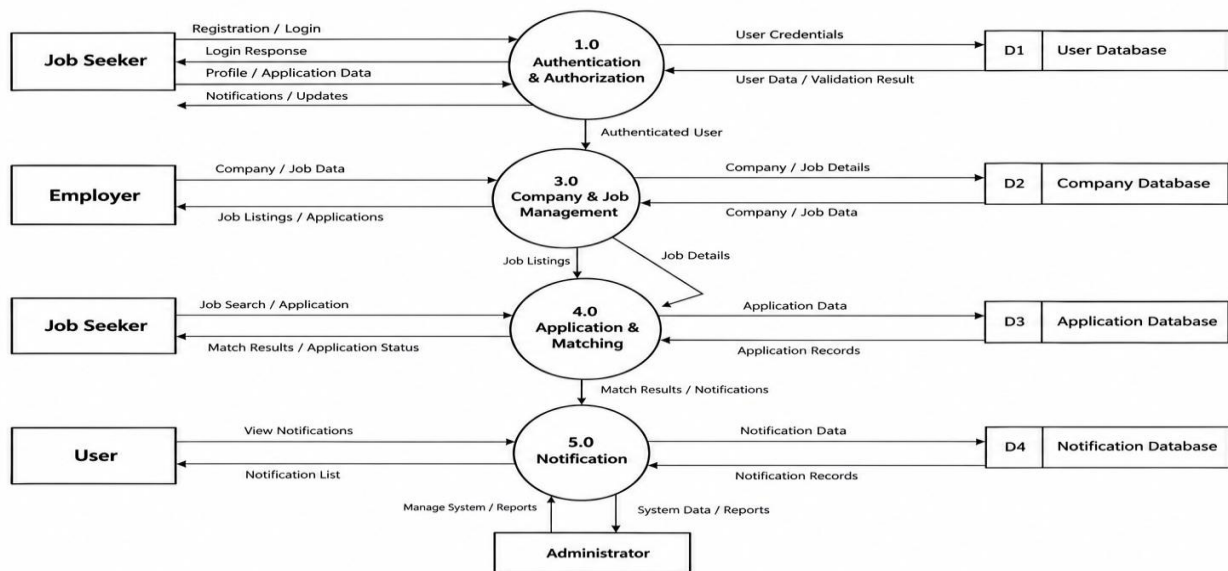
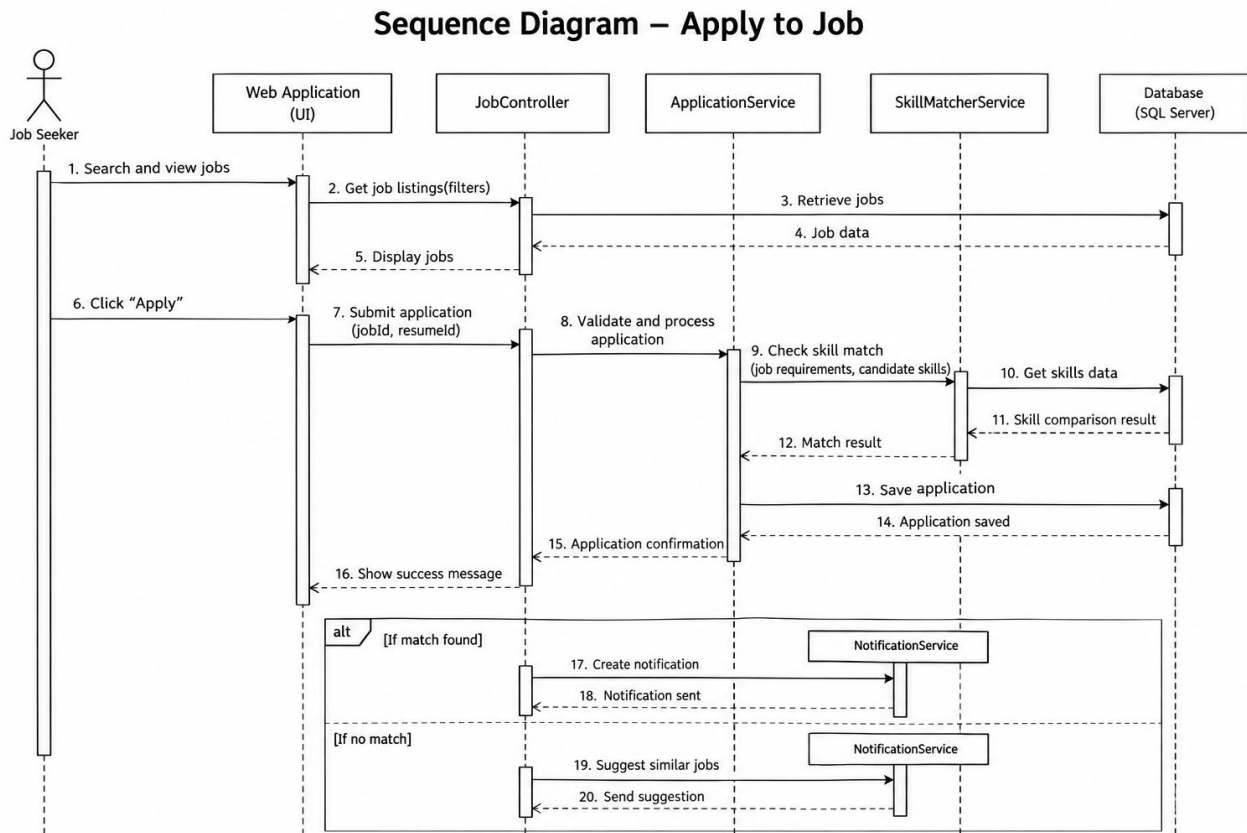


Figure 6.12: Level-1 Data Flow Diagram

**Table 6.3: Major Data Flow Processes**

| Process                  | Input                             | Processing                            | Output                           |
|--------------------------|-----------------------------------|---------------------------------------|----------------------------------|
| Authentication           | Registration/login data           | Validate identity                     | Authenticated session            |
| Profile Management       | Profile, resume, skills           | Validate and save                     | Updated profile                  |
| Company & Job Management | Company/job information           | Validate and store                    | Company/job records              |
| Application & Matching   | Application and skill information | Create application and compare skills | Application/matching information |
| Notification             | System events                     | Create and store notification         | User notification                |
| Administration           | Administrative request            | Perform authorized operation          | Administrative result            |

## 6.6 Sequence Diagram for SmartHire:



**Figure 6.15: Sequence Diagram for Applying Job**

## 6.7 Class Diagram for SmartHire:

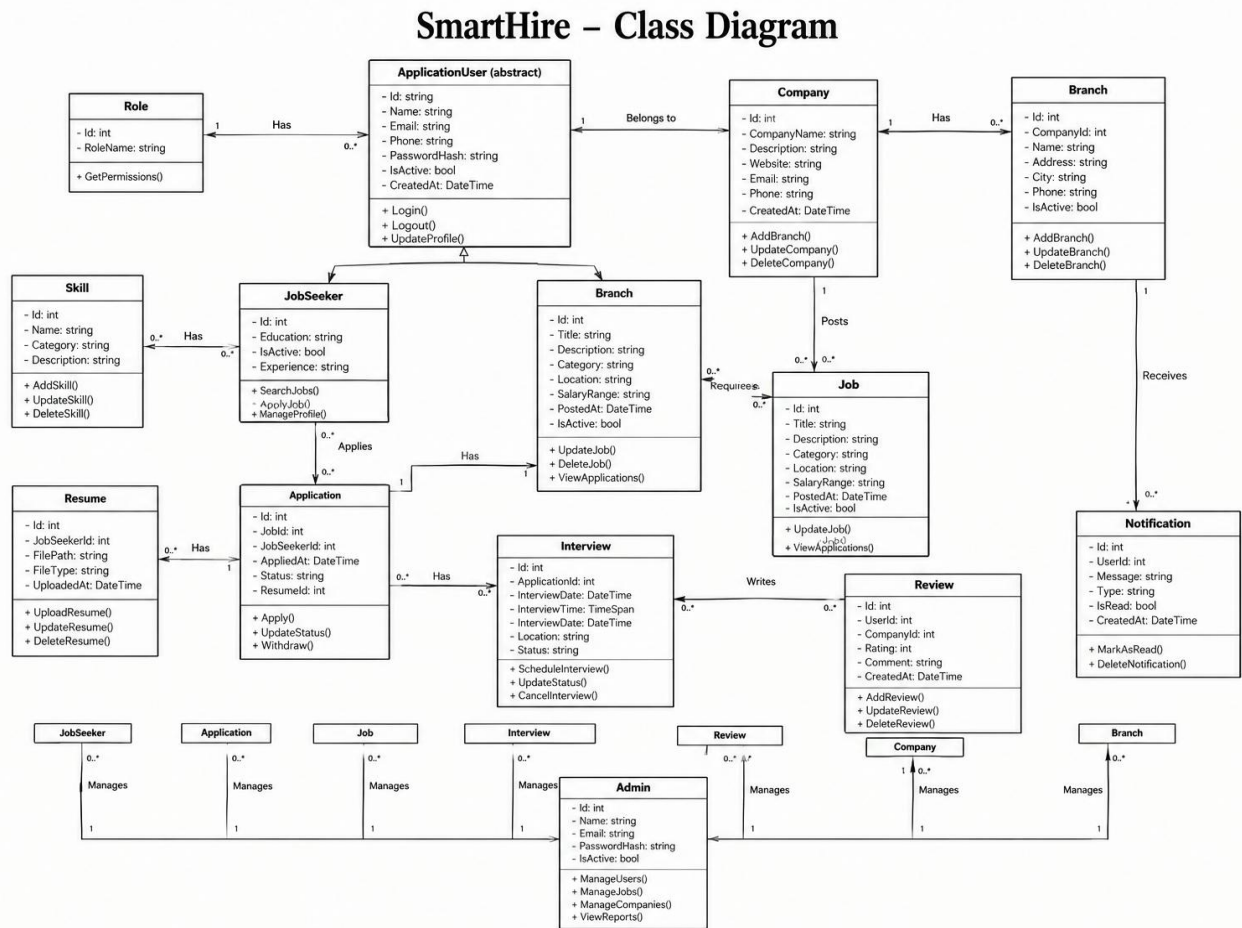


Figure 6.16: Class Diagram for

## 6.8 Entity Relationship and Database Structure

SmartHire uses a relational database with Microsoft SQL Server and Entity Framework Core. The system follows the Code First approach, where application models define the database structure and migrations are used to create and update the database.

The database includes ASP.NET Core Identity tables along with application-specific entities for users, companies, employers, job seekers, jobs, skills, resumes, applications, notifications, interviews, and related activities.

### 6.8.1 ApplicationUser

ApplicationUser represents the authenticated user account in SmartHire. It works with ASP.NET Core Identity to provide authentication and role-based access to the system.

### **6.8.2 Company**

Company represents an organization using SmartHire for recruitment. A company can have multiple branches and can be associated with employers, jobs, followers, and reviews.

### **6.8.3 Branch**

Branch represents a branch or organizational unit of a company. A company may have multiple branches, and employer accounts can be associated with a specific branch.

Relationship:

Company → Branch → Employer

### **6.8.4 Employer**

Employer represents a company-side user who participates in recruitment activities. It is associated with an ApplicationUser, Company, and Branch. The employer's position and approval information help distinguish owners from other company employees.

### **6.8.5 JobSeeker**

JobSeeker represents a candidate using SmartHire to search and apply for jobs. A job seeker can maintain skills, resumes, applications, saved jobs, and other profile information.

### **6.8.6 Job**

Job represents a vacancy posted by an authorized employer. It stores key information such as job title, description, location, salary, category, deadline, and status. A job can also have multiple required skills and applications.

### **6.8.7 Skill**

Skill represents a reusable skill that can be associated with both jobs and job seekers. It forms the basis of the system's skill-based matching process.

### **6.8.8 JobSkill**

JobSkill connects a job with its required skills. This allows a job to have multiple required skills while allowing the same skill to be associated with multiple jobs.

Relationship:

Job → JobSkill → Skill

### **6.8.9 JobSeekerSkill**

JobSeekerSkill connects a job seeker with their skills. This structure allows multiple skills to be associated with a job seeker and supports comparison with job requirements.

Relationship:

JobSeeker → JobSeekerSkill → Skill

### 6.8.10 Resume

Resume stores resume information associated with a job seeker. A job seeker can use a selected resume when applying for a job.

Relationship:

JobSeeker → Resume → JobApplication

### 6.8.11 JobApplication

JobApplication represents an application submitted by a job seeker for a specific job. It connects the Job Seeker, Job, and Resume and stores application information such as status and application date.

Relationship:

JobSeeker + Job + Resume → JobApplication

### 6.8.12 Notification

**Table 6.4: Major Database Entities of SmartHire**

| Entity           | Main Purpose                     | Major Relationship               |
|------------------|----------------------------------|----------------------------------|
| ApplicationUser  | User identity and authentication | JobSeeker/Employer               |
| Company          | Organization information         | Branch, Employer, Job            |
| Branch           | Company branch/unit              | Company, Employer                |
| Employer         | Company-side user                | User, Company, Branch, Job       |
| JobSeeker        | Candidate profile                | User, Skill, Resume, Application |
| Job              | Recruitment vacancy              | Employer, Skill, Application     |
| Skill            | Reusable skill information       | Job and JobSeeker                |
| JobSkill         | Job-skill relationship           | Job and Skill                    |
| JobSeekerSkill   | Candidate-skill relationship     | JobSeeker and Skill              |
| Resume           | Candidate resume                 | JobSeeker and Application        |
| JobApplication   | Candidate's application          | Job, JobSeeker, Resume           |
| Notification     | User notification                | User, Job/Application            |
| Interview        | Interview information            | Recruitment/Application          |
| InterviewMessage | Interview communication          | Interview                        |
| CompanyFollow    | Company-follow relationship      | JobSeeker and Company            |
| CompanyReview    | Company review                   | JobSeeker and Company            |

### 6.8.17 Main Database Relationships

SmartHire's database is built around a number of key relationships that connect users, companies, jobs, and applications together.

Every user account (ApplicationUser) is linked to a role-specific profile, either a JobSeeker or an Employer, depending on how the account was registered. This keeps the login and identity information separate from the details specific to each role.

On the organization side, a Company can have several Branches, and each Branch can have its own Employer accounts. This structure allows a large company to manage recruitment across different locations while still keeping everything under one company profile.

An Employer, once verified, is able to create and manage Job postings. Each Job can be linked to multiple Skills through a JobSkill table, which records what skills are required for that role. In the same way, a JobSeeker's own skills are stored through a JobSeekerSkill table, connecting candidates to the Skill entity.

When a JobSeeker applies for a position, this creates a JobApplication record, which connects the Job and the JobSeeker together. A JobApplication can also be linked to a Resume, so the employer can see which resume was submitted for that particular application. From there, a JobApplication may generate a Notification (for example, when the status changes) and may also lead to an Interview if the employer decides to move forward with the candidate.

### 6.8.18 Referential Integrity

Because so many tables in SmartHire are connected through foreign keys, maintaining referential integrity is essential. A JobApplication, for instance, must always point to a real, existing Job and JobSeeker—it cannot reference records that don't exist. The same applies to Employers, who must be linked to a valid Company and Branch.

To enforce this, SmartHire relies on Entity Framework Core's relationship configuration along with SQL Server's own foreign-key constraints. One issue that came up during development was that some entities have more than one possible delete path leading to the same table, which SQL Server does not allow with cascading deletes by default. To avoid this conflict, certain relationships were configured with DeleteBehavior.Restrict instead of cascade delete, meaning a parent record cannot be removed while related child records still exist.

### 6.8.19 Database Structure Overview

To simplify the database structure, the tables are grouped according to their functional roles:

1. **Identity:**AspNetUsers, AspNetRoles, and related Identity tables
2. **Organization:** Companies, Branches, Employers
3. **Candidate:** JobSeekers, Resumes, Skills, Education, Experience, Certifications, Projects
4. **Recruitment:** Jobs, JobSkills, JobApplications, SavedJobs

5. **Communication:** Notifications, Interviews, InterviewMessages
6. **Company Engagement:** CompanyFollows, CompanyReviews
7. **AI/Analysis:** AIAAnalyses for AI-based analysis and matching

This structure clearly represents the complete recruitment workflow, from user registration and job applications to AI analysis, communication, and company engagement.

## Entity Relationship Diagram:

### SmartHire – Entity Relationship Diagram

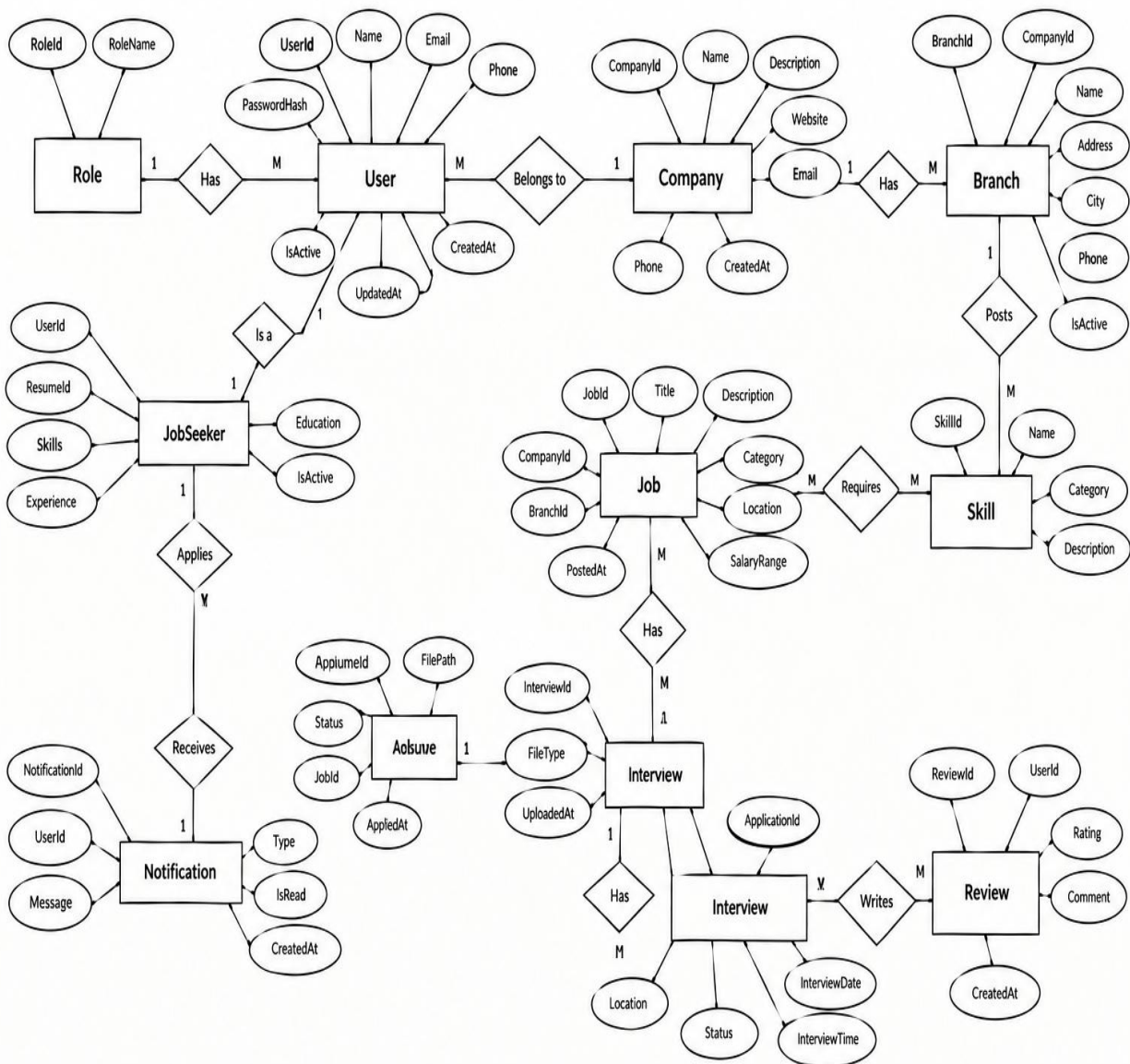


Figure 6.17: SmartHire Entity Relationship Diagram (ERD)

## **Chapter 9**

### **Conclusion**

## **9.1 Preface**

Online recruitment systems have become important for connecting job seekers and employers efficiently. SmartHire – AI-Powered Smart Job Search Management System provides a centralized platform for job searching, profile and resume management, applications, job posting, and recruitment activities.

Developing SmartHire allowed me to apply my academic knowledge in a practical software environment. It strengthened my skills in database design, MVC architecture, programming, testing, debugging, and user-interface development.

## **9.2 Practicum and Its Value**

The main purpose of the practicum was to develop a web-based recruitment system that simplifies the hiring process for both job seekers and employers.

For job seekers, SmartHire supports profile and skill management, resume upload, job search, applications, application tracking, skill matching, and notifications. For employers, it provides company and employee management, job posting, application review, and recruitment communication.

The practicum provided practical experience with C#, ASP.NET Core MVC, Entity Framework Core, SQL Server, Razor Views, HTML, CSS, Bootstrap, authentication, authorization, and software testing. It also improved my understanding of requirement analysis, database and UML modeling, debugging, and real-world software development.

## **9.3 Ethical Considerations**

Ethical considerations were taken into account during the development of the SmartHire system because the platform handles personal information such as user profiles, resumes, skills, job applications, and recruitment-related communication. Access to these resources is controlled through authentication and role-based authorization to reduce unauthorized access.

The system is designed to use candidate information only for relevant recruitment activities. Skill matching is based on available job requirements and candidate skills, and the result is intended to support the recruitment process rather than completely replace the employer's judgment. The current matching process is rule-based, which also makes the matching result more transparent and understandable.

## **9.4 Future Work**

The system can be improved in the future by adding more intelligent and automated recruitment features. Possible future improvements include:

- Advanced AI-based job and candidate recommendation.
- Semantic skill matching using modern NLP techniques.

- Automatic resume information extraction.
- AI-assisted candidate ranking and shortlisting.
- More advanced recruitment analytics and statistical reports.
- Email and SMS notifications.
- Improved interview management and online interview support.
- Advanced search and filtering options.
- Mobile application support.
- Cloud deployment and improved scalability.
- Enhanced security, monitoring, backup, and system administration features.

These improvements can make SmartHire more intelligent, scalable, and suitable for wider real-world use. Similar to the reference report, future development can focus on adding additional features based on user feedback and requirements while continuously improving and maintaining the system.

## 9.5 Conclusion

In conclusion, SmartHire – AI-Powered Smart Job Search Management System provides a centralized and structured solution for managing modern recruitment activities. The system connects job seekers, employers, and administrators through a web-based platform and supports important activities such as job searching, profile and resume management, job posting, application management, skill matching, notifications, interview management, and recruitment communication.

The development of this project allowed me to gain practical experience in designing and implementing a complete web application using ASP.NET Core MVC, C#, Entity Framework Core, and Microsoft SQL Server. It also strengthened my understanding of software engineering concepts, database management, authentication, authorization, UI development, testing, debugging, and project documentation.

The practicum has been an important part of my academic and professional development. It helped me understand how theoretical concepts are applied in real software projects and improved my ability to analyze problems, develop solutions, and work systematically. Overall, SmartHire achieved its primary objectives and provides a strong foundation for further development into a more advanced and intelligent recruitment platform.